

100+ Free Courses. Six Domains. One Guide.

Every high-quality free course, certificate programme, and learning resource – **Verified, clickable, and ready to use.**

Computer Science

AI & Machine Learning

Data Science

Cybersecurity

Cloud & IT

GRC & AI Governance

Harvard · MIT · Stanford · Google · IBM · Anthropic

Harvard



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Six Domains. Zero Cost.

Six sections. Direct links. Zero cost.

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HOW TO READ THIS GUIDE

CERTIFICATE CLASSIFICATION

FREE CERT

Course content and certificate are both free. No paywall at the end.

CONTENT FREE

All materials free to access. Certificate optional and may carry a small fee.

LEVEL CLASSIFICATION

Beginner

No prior knowledge needed.

Intermediate

Some foundational knowledge helpful.

Advanced

Solid technical foundation required.

Computer Science Fundamentals

Algorithmic thinking, programming fluency, data structures, and systems knowledge. The mental models everything else depends on.

COURSE	INSTITUTION	LEVEL	CERT	WHAT IT COVERS	LINK
CS50x: Intro to Computer Science	Harvard	Beginner	FREE CERT	Algorithms, data structures, C, Python, SQL, web dev. The most-enrolled CS course on Earth. Free Harvard certificate on completion.	cs50.harvard.edu/x
CS50 Python	Harvard	Beginner	FREE CERT	Python from functions through OOP, files, libraries, and testing. Free Harvard certificate — no edX account needed.	cs50.harvard.edu/python
CS50 Web Programming	Harvard	Intermediate	FREE CERT	Full-stack web: Django, JavaScript, React, SQL, security. Free Harvard certificate.	cs50.harvard.edu/web
CS50 SQL	Harvard	Beginner	FREE CERT	Database design, SQL querying, normalisation, transactions. Free Harvard certificate.	cs50.harvard.edu/sql
Intro to CS and Programming (6.0001)	MIT OCW	Beginner	CONTENT FREE	Python from scratch, computational thinking, algorithms. Full MIT lectures and problem sets — no signup needed.	ocw.mit.edu
Introduction to Algorithms (6.006)	MIT OCW	Intermediate	CONTENT FREE	Sorting, graphs, dynamic programming, hashing. Full lecture videos and problem sets free.	ocw.mit.edu
The Missing Semester of CS Education	MIT	Beginner	CONTENT FREE	Shell, Vim, Git, debugging, SSH, security — practical tools few CS programmes teach. Lecture videos and notes free.	missing.csail.mit.edu
Introduction to Internet of Things	Stanford	Beginner	CONTENT FREE	What IoT is, how connected devices work, sensing, actuation, networking, and embedded systems. No prior experience needed.	Stanford Online
Algorithms, Part I & II	Princeton	Intermediate	CONTENT FREE	Sorting, graphs, string processing — by Sedgewick. The definitive algorithms resource. Lecture videos on YouTube.	algs4.cs.princeton.edu

COURSE	INSTITUTION	LEVEL	CERT	WHAT IT COVERS	LINK
freeCodeCamp Certifications	freeCodeCamp	All Levels	FREE CERT	Responsive Web Design, JS Algorithms, Front End Libraries, Back End, Python — all certifications and all certificates 100% free.	freecodecamp.org/learn
Structure and Interpretation of Computer Programs (6.001)	MIT OCW	Advanced	CONTENT FREE	Functional programming, abstraction, interpreters. A legendary course in CS history. Full video lectures free.	ocw.mit.edu

CURATOR'S PICK

Harvard CS50x at cs50.harvard.edu/x is the single best starting point in tech education anywhere on the internet. Rigorous, project-based, and free — including the certificate. If you do one thing from this guide, start here.

Artificial Intelligence & Machine Learning

From foundational ML to LLMs, AI agents, and generative AI. Taught by the researchers who built these systems — and now also the companies deploying them.

COURSE	SOURCE	LEVEL	CERT	WHAT IT COVERS	LINK
Machine Learning Crash Course	Google	Beginner	FREE CERT	Refreshed 2024 edition: loss, gradient descent, classification, neural nets, LLM intro. Interactive browser coding. Free module badges.	developers.google.com/ml
Elements of AI — Introduction to AI	Univ. Helsinki	Beginner	FREE CERT	What AI is, what it can't do, societal impact — no maths needed. 2 million+ students. Free digital certificate on completion.	elementsofai.com
Building AI	Univ. Helsinki	Intermediate	FREE CERT	Actual ML algorithms: regression, neural nets, Bayesian models. Requires basic Python. Free certificate on completion.	buildingai.elementsofai.com
Practical Deep Learning for Coders	fast.ai	Intermediate	CONTENT FREE	Code-first deep learning with PyTorch. CV, NLP, tabular data, deployment. By Jeremy Howard. Fully free, no account needed.	course.fast.ai
MIT 6.S191: Intro to Deep Learning	MIT	Intermediate	CONTENT FREE	CNNs, RNNs, transformers, GANs, RL, LLMs. Annual updated lectures and labs free on YouTube and course website.	introtodeeplearning.com
CS229: Machine Learning	Stanford	Advanced	CONTENT FREE	The original graduate ML course: regression, SVMs, neural nets, EM, RL. Lecture notes and videos free at cs229.stanford.edu .	cs229.stanford.edu
CS231n: Deep Learning for Computer Vision	Stanford	Advanced	CONTENT FREE	CNNs, object detection, segmentation, generative models. Lecture notes and YouTube recordings free. The definitive CV course.	cs231n.github.io
CS224N: NLP with Deep Learning	Stanford	Advanced	CONTENT FREE	Transformers, LLMs, BERT, GPT, alignment — by Chris Manning. 2024 lecture videos free on YouTube.	web.stanford.edu/cs224n

COURSE	SOURCE	LEVEL	CERT	WHAT IT COVERS	LINK
LLM Course (Transformers, Fine-tuning)	Hugging Face	Intermediate	CONTENT FREE	NLP fundamentals through LLM fine-tuning and deployment using the Hugging Face ecosystem. Fully free, no account needed.	huggingface.co/learn/llm-course
AI Agents Course	Hugging Face	Intermediate	FREE CERT	Build AI agents with smol-agents, LlamaIndex, LangGraph. Hands-on in Hugging Face Spaces. Free certificate on completion.	huggingface.co/learn/agents-course
5-Day Gen AI Intensive	Kaggle / Google	Intermediate	CONTENT FREE	Foundational LLM models, embeddings, agents, and MLOps for GenAI — a structured 5-day programme with notebooks and whitepapers. Free access, no subscription.	kaggle.com/learn-guide/5-day-genai
5-Day AI Agents Intensive	Kaggle / Google	Intermediate	CONTENT FREE	Agentic AI: tool use, function calling, multi-agent systems, and evaluation — 5-day structured deep-dive with live notebooks.	kaggle.com/learn-guide/5-day-agents
Anthropic Academy — 13 Free AI Courses	Anthropic	All Levels	FREE CERT	13 official courses covering Claude 101, AI Fluency, Claude API, Claude Code, MCP, Agent Skills, and enterprise deployment. Every course is free with an official Anthropic certificate. Launched March 2026 — directly from the makers of Claude.	anthropic.skilljar.com
DeepLearning.AI Short Courses (25+)	DeepLearning.AI	Intermediate	CONTENT FREE	1-3 hour practical courses on prompt engineering, RAG, LangChain, fine-tuning, agents — co-created with OpenAI, Anthropic, Google, AWS. All free on platform, no credit card.	deeplearning.ai/short-courses
MIT 6.034: Artificial Intelligence	MIT OCW	Intermediate	CONTENT FREE	Search, knowledge representation, ML, logic — classic Patrick Winston lectures. Full videos free on MIT OCW.	ocw.mit.edu

Data Science & Analytics

Python, SQL, statistics, visualisation, and applied analytics. The skills most data and analytics roles require from day one.

COURSE	SOURCE	LEVEL	CERT	WHAT IT COVERS	LINK
Kaggle Learn — Full Course Library	Kaggle (Google)	All Levels	FREE CERT	Python, Pandas, Data Visualisation, SQL, Intro ML, Intermediate ML, Feature Engineering, AI Ethics — all courses and all certificates free. In-browser coding, no setup.	kaggle.com/learn
Python for Data Science	IBM Cognitive Class	Beginner	FREE CERT	Python fundamentals, NumPy, Pandas, APIs, web scraping. Free badge on IBM Cognitive Class — no Coursera account needed.	cognitiveclass.ai
Data Analysis with Python	IBM Cognitive Class	Beginner	FREE CERT	Pandas, NumPy, data wrangling, EDA, and model evaluation. Free digital badge — no subscription required.	cognitiveclass.ai
Prompt Engineering for Everyone	IBM Cognitive Class	Beginner	FREE CERT	Foundations of prompt engineering: instruction design, context setting, few-shot examples, chain-of-thought techniques. Free badge on IBM Cognitive Class.	cognitiveclass.ai
Machine Learning with Python	IBM Cognitive Class	Intermediate	FREE CERT	Regression, classification, clustering, model evaluation — scikit-learn and Python. Free Cognitive Class certificate on completion.	cognitiveclass.ai
Data Science: R Basics	Harvard	Beginner	CONTENT FREE	Introduction to R for data analysis and visualisation. First course in Harvard's Data Science series. Lectures available free.	pll.harvard.edu
Statistics for Applications (18.650)	MIT OCW	Intermediate	CONTENT FREE	Probability, estimation, hypothesis testing, regression — graduate statistics with full MIT lecture videos and problem sets.	ocw.mit.edu
CS50 SQL	Harvard	Beginner	FREE CERT	Database design, SQL, normalisation, transactions, optimisation. Practical project-based. Free Harvard certificate.	cs50.harvard.edu/sql

Cybersecurity

From SOC analyst foundations to web application exploitation and network defence. The fastest-growing technical field — and uniquely accessible to self-directed learners.

COURSE / RESOURCE	SOURCE	LEVEL	CERT	WHAT IT COVERS	LINK
ISC2 Certified in Cybersecurity (CC)	ISC2	Beginner	FREE CERT	Official ISC2 CC training AND free exam for candidates. Security principles, access controls, network security, incident response, cryptography. Globally recognised entry-level credential.	isc2.org/candidate
CS50 Cybersecurity	Harvard	Beginner	FREE CERT	Passwords, phishing, malware, privacy, cryptography, web security. CS50 pedagogy. Free Harvard certificate — no experience needed.	cs50.harvard.edu/cybersecurity
Web Security Academy	PortSwigger	All Levels	CONTENT FREE	190+ interactive labs: SQL injection, XSS, CSRF, authentication, SSRF, IDOR. By the creators of Burp Suite. The best free web security training available.	portswigger.net/web-security
TryHackMe Free Paths	TryHackMe	Beginner	CONTENT FREE	Pre-Security and SOC Level 1: networking, Linux, web hacking, SIEM — gamified, browser-based, no software required. Free tier covers substantial foundational content.	tryhackme.com/paths
Introduction to Cybersecurity	Cisco NetAcad	Beginner	FREE CERT	Threat landscape, malware types, attack vectors, defence basics, career paths. Free with digital badge on completion.	netacad.com
Networking Basics	Cisco NetAcad	Beginner	FREE CERT	OSI model, IP addressing, switching, routing, wireless. Free with digital badge. Foundational for any security or cloud role.	netacad.com
IBM Cognitive Class — Cybersecurity Courses	IBM Cognitive Class	All Levels	FREE CERT	Three free cybersecurity courses: Cybersecurity Fundamentals, Cyber Threat Intelligence, and Network Security — all with free digital badges. No subscription needed.	cognitiveclass.ai/cybersecurity
IBM SkillsBuild Cybersecurity Path	IBM SkillsBuild	Beginner	FREE CERT	Network security, endpoint protection, incident response. Free digital badge credentials via IBM SkillsBuild.	skillsbuild.org

Cloud Computing & IT

AWS, Azure, Google Cloud, Linux, networking, and DevOps. The infrastructure layer every modern tech role depends on.

COURSE	SOURCE	LEVEL	CERT	WHAT IT COVERS	LINK
AWS Skill Builder — 600+ Free Courses	Amazon AWS	All Levels	FREE CERT	Every AWS domain: compute, storage, ML, security, networking, DevOps. 600+ free courses, free digital badges. No credit card, no subscription.	skillbuilder.aws
AWS Cloud Practitioner Essentials	Amazon AWS	Beginner	FREE CERT	AWS services, architecture, billing, security, Shared Responsibility Model. Direct prep for the Cloud Practitioner exam. Free on Skill Builder.	skillbuilder.aws
Microsoft Learn — AZ-900 Full Path	Microsoft	Beginner	CONTENT FREE	Full official AZ-900 learning path: cloud concepts, Azure services, pricing, compliance, governance. 100% free on Microsoft Learn, no account needed.	learn.microsoft.com
Microsoft Learn — Full Catalogue	Microsoft	All Levels	CONTENT FREE	Hundreds of free learning paths: Azure, AI, DevOps, security, Power Platform. All content fully free with no subscription. Certification exams carry separate fees.	learn.microsoft.com
Google Cloud Skills Boost	Google	Beginner	FREE CERT	Hands-on labs in real GCP environments. Kubernetes, BigQuery, ML, security. GEAR community members get 35 free monthly lab credits. Skill badges shareable on LinkedIn.	cloudskillsboost.google
Introduction to Linux (LFS101)	Linux Foundation	Beginner	CONTENT FREE	Official Linux Foundation course. Command line, file systems, package management, shell scripting, and networking. 14 weeks, fully self-paced, directly on the Linux Foundation site.	linuxfoundation.org
Linux Essentials	Cisco NetAcad	Beginner	FREE CERT	Linux command line, file systems, permissions, networking — from Cisco's Networking Academy. Free digital badge on completion.	netacad.com

COURSE	SOURCE	LEVEL	CERT	WHAT IT COVERS	LINK
Docker Essentials	IBM Cognitive Class	Intermediate	FREE CERT	Containerisation, Docker fundamentals, Kubernetes orchestration, cloud-native development. Free digital badge on IBM Cognitive Class.	cognitiveclass.ai
Intro to DevOps and SRE	Linux Foundation	Intermediate	CONTENT FREE	DevOps culture, CI/CD pipelines, infrastructure as code, monitoring, SRE principles. Free Linux Foundation course — 40+ hours. Certification exam is separate.	linuxfoundation.org

CURATOR'S PICK

AWS Skill Builder at skillbuilder.aws is the most underused free resource in tech. 600+ courses, real AWS exposure, free badges — no subscription. Start with the Cloud Practitioner Essentials learning plan. Zero to cloud-ready.

GRC & AI Governance

Governance, risk, compliance, privacy, AI governance, and information security frameworks. Certifications, courses, and training from the organisations setting the standards.

COURSE / CERTIFICATION	SOURCE	LEVEL	CERT	WHAT IT COVERS	LINK
ISC2 Certified in Cybersecurity (CC)	ISC2	Beginner	FREE CERT	Security principles, risk management, access controls, governance – foundational GRC knowledge with a globally recognised credential. Free training and free exam.	isc2.org/candidate
NIST RMF: 4 Introductory Online Courses	NIST	Beginner	FREE CERT	NIST Risk Management Framework: prepare, categorise, select, implement, assess, authorise, monitor. Official free training from NIST directly.	csrc.nist.gov
OneTrust AI Governance Certification	OneTrust	Intermediate	FREE CERT	AI governance frameworks, risk assessment, bias and fairness, EU AI Act alignment. Free certification from a leading GRC technology vendor.	onetrust.com/certifications
OneTrust Tech Risk & Compliance Cert	OneTrust	Intermediate	FREE CERT	Technology risk management, compliance programme design, control frameworks, audit readiness. Free OneTrust certification.	onetrust.com/certifications
OneTrust Third-Party Management Cert	OneTrust	Intermediate	FREE CERT	Vendor risk management, third-party due diligence, contract management, continuous monitoring. Free OneTrust certification.	onetrust.com/certifications
AI Security and Governance Certification	Securiti.ai	Intermediate	FREE CERT	AI governance, AI risk management, responsible AI, global AI laws, EU AI Act alignment. Free certification from Securiti.ai Education platform.	education.securiti.ai
DSPM Fundamentals Certification	Securiti.ai	Beginner	FREE CERT	Data Security Posture Management: data discovery, classification, risk quantification, AI security governance. Free Securiti.ai certification.	education.securiti.ai

COURSE / CERTIFICATION	SOURCE	LEVEL	CERT	WHAT IT COVERS	LINK
PrivacyOps Certification	Securiti.ai	Intermediate	FREE CERT	Privacy operations, global data privacy laws, DSAR workflows, consent management, and privacy engineering. Free Securiti.ai certification.	education.securiti.ai
SOC 2 Master Implementer Certification	Scytale	Intermediate	FREE CERT	SOC 2 Trust Services Criteria, control implementation, audit readiness, evidence collection. Free certification from Scytale.	scytale.ai/free-soc2-training
GDPR Training Course	Measured Collective	Beginner	FREE CERT	GDPR principles, lawful bases, data subject rights, breach notification, DPO obligations — UK and EU GDPR. Free certificate on assessment completion.	measuredcollective.com
DPDPA India Certificate Course	DPDPA.com	Beginner	FREE CERT	India's Digital Personal Data Protection Act: obligations, data fiduciary rules, consent framework, enforcement. Free certificate course.	dpdpa.com
CISO Workshop Training	Microsoft	Intermediate	CONTENT FREE	Security strategy, risk management, Zero Trust architecture, compliance posture — delivered by Microsoft to security and executive leaders. Free workshop content on Microsoft Learn.	learn.microsoft.com
Introduction to Zero Trust	Microsoft	Beginner	CONTENT FREE	Zero Trust principles, identity, device, network, application and data security layers, best practice frameworks. Free on Microsoft Learn.	learn.microsoft.com
Qualys Policy Compliance Learning Path	Qualys	Intermediate	FREE CERT	Policy compliance scanning, CIS benchmarks, control validation, remediation workflows — using Qualys platform. Free learning path with certification.	qualys.com/training
Qualys PCI Compliance Learning Path	Qualys	Intermediate	FREE CERT	PCI-DSS requirements, vulnerability scanning, web app firewalls, and compliance reporting via Qualys. Free learning path with certification.	qualys.com/training

COURSE / CERTIFICATION	SOURCE	LEVEL	CERT	WHAT IT COVERS	LINK
How to Break into GRC	SimplyCyber	Beginner	CONTENT FREE	Mindset, methods, and skills for entering GRC as a career — practical guidance for early-career professionals and career changers.	SimplyCyber Academy
How to Grow Your GRC Consultancy	Advisera	Advanced	CONTENT FREE	Advanced course on building and scaling a cybersecurity, ISO, and AI consultancy. Practical business and delivery guidance.	advisera.com

CURATOR'S PICK

Start with **ISC2 CC** for a credentialled GRC foundation, then layer **NIST RMF** for frameworks and **OneTrust** for hands-on platform knowledge. For AI governance specifically, the **Securiti.ai AI Governance Certification** is the most comprehensive free credential available in the field right now.

Direct Access by Domain

Every platform, direct URL, and access note in one place.

Harvard CS50 Suite — cs50.harvard.edu

- CS50x (CS fundamentals): cs50.harvard.edu/x
- CS50 Python: cs50.harvard.edu/python
- CS50 Web: cs50.harvard.edu/web
- CS50 AI: cs50.harvard.edu/ai
- CS50 SQL: cs50.harvard.edu/sql
- CS50 Cybersecurity: cs50.harvard.edu/cybersecurity
- **All certificates free on completion — no edX account needed**

MIT OpenCourseWare — ocw.mit.edu

- All MIT courses free, no signup, no start dates: ocw.mit.edu
- 6.0001 Python: search "6.0001" on OCW
- 6.006 Algorithms: search "6.006" on OCW
- 6.034 AI: search "6.034" on OCW
- Missing Semester: missing.csail.mit.edu
- 6.S191 Deep Learning: introtodeeplearning.com

Google Platforms

- ML Crash Course: developers.google.com/ml
- Cloud Skills Boost: cloudskillsboost.google
- Kaggle Learn (all certs free): kaggle.com/learn
- Kaggle 5-Day GenAI: kaggle.com/learn-guide/5-day-genai
- Kaggle 5-Day Agents: kaggle.com/learn-guide/5-day-agents

Amazon AWS

- Skill Builder (600+ free): skillbuilder.aws
- No credit card required for free tier
- Free digital badges on 100+ courses
- Start: Cloud Practitioner Essentials learning plan

Anthropic & AI Specialist Platforms

- Anthropic Academy (13 free courses): anthropic.skilljar.com
- fast.ai (no account): course.fast.ai
- DeepLearning.AI short courses: deeplearning.ai/short-courses
- Hugging Face Learn: huggingface.co/learn
- Elements of AI (free cert): elementsofai.com

IBM Platforms

- Cognitive Class: cognitiveclass.ai
- SkillsBuild: skillsbuild.org
- 600+ free courses, 60+ with free digital badges
- No Coursera subscription needed on either platform

Cybersecurity Specialist Platforms

- ISC2 CC (free exam): isc2.org/candidate
- PortSwigger Web Security: portswigger.net/web-security
- TryHackMe: tryhackme.com
- Cisco NetAcad: netacad.com

GRC & AI Governance Platforms

- NIST RMF courses: csrc.nist.gov/rmf
- OneTrust certifications: onetrust.com/certifications
- Securi.ai certifications: securiti.ai
- SOC 2 Free Training (Scytale): scytale.ai/free-soc2-training
- Qualys Training: qualys.com/training
- Microsoft CISO Workshop: learn.microsoft.com

All The Best Start Learning Now

**Up Next - AI Projects that Solve
Real-World Problems**